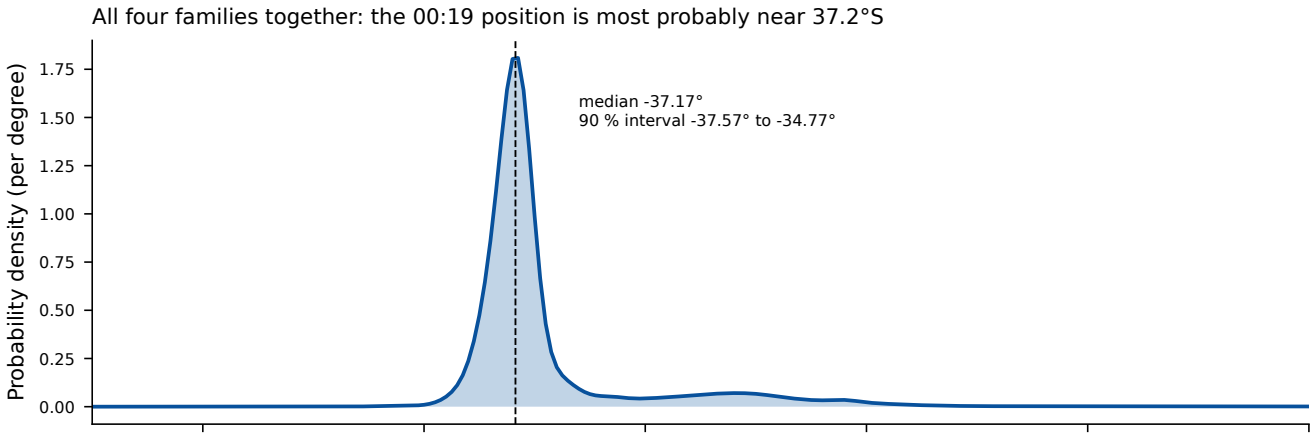
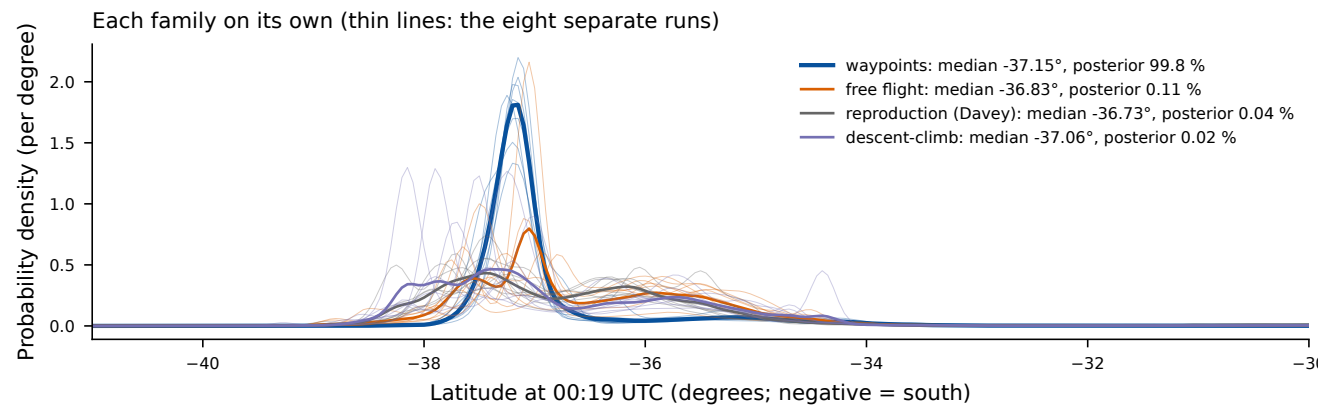


b



Simple version (ASD-STE100): The curves show where the aircraft probably was at 00:19 UTC, along the seventh satellite arc. Panel a combines the four groups of possible early flights. Each group has the same prior probability, and the data then give each group its share. The waypoint group has almost all of the probability, so panel a is almost the same as the waypoint curve in panel b. The thin lines are the eight separate runs. Where they do not agree, the result is less sure.

Technical version: Core run I, 12 Oct 2026; 8 seeds per family (7M particles per seed for free and waypoints, 3.5M for descent-climb and reproduction). Densities of 00:19 latitude from summary.json (0.1° smoothing), core 00:19 treatment: Davey both-BTO, no BFO. Mixture weights are family posteriors at equal prior from pooled log-evidence (waypoints −129.00, free −135.78, reproduction −136.73, descent-climb −137.46). Split-half over 35 balanced partitions: waypoints 0.916, free 0.830, reproduction 0.891, descent-climb 0.794; none reaches the 8-seed floor 0.924. Speed rule for free, waypoints and descent-climb: early Mach 0.45–0.87 within 210–330 KCAS to 18:40, cruise 0.70–0.87 within 210–330 KCAS; reproduction Davey Mach 0.73–0.84. One-engine phase (s7 + s8) provisional.